



SRI SATHYA SAI INSTITUTE OF HIGHER LEARNING

(Deemed to be University)

Department of Mathematics and Computer Science

CIE III - SOLUTIONS

Designing Tables and Graphs

UDSC-402: Data Analysis and Visualization

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Date: 06-10-2026

Q1. State whether the following statements are True or False. [4 × 1 = 4 Marks]

(i) A single set of values occupying one column is, by definition, a table.

FALSE. It is the arrangement in columns and rows that characterises a table. A single set of values occupying one column or one row is merely a list.

(ii) Months of the year formally represent an interval relationship rather than an ordinal one.

TRUE. Months form a sequential series of numerical ranges subdividing a larger range (a time series). Although months are not equally long, they are treated as approximately equal intervals for reporting.

(iii) Hue is perceived quantitatively and may therefore be used to encode quantitative values accurately.

FALSE. Different hues are perceived only as categorically different, never as more or less. Only length and 2-D position encode quantity accurately.

(iv) A scale break is acceptable in a graph provided it is clearly marked for the reader.

FALSE. Scale breaks mislead even when marked — a break between 400 and 8,000 badly under-represented a dramatic rise in influenza cases. The remedy is to use two separate graphs.

Q2. Fill in the blanks. [4 × 1 = 4 Marks]

(i) The amount of ink that presents information compared to the total ink used is called the _____ ratio.

data-ink ratio (Tufte). The object is not to remove all non-data ink, but to reduce it to no more than is necessary.

(ii) Working memory is severely limited and can hold only _____ or _____ chunks at a time.

three or four chunks. This is why a legend with ten colours or symbols fails.

(iii) A table in which categorical items are laid out both across the columns and down the rows is called a _____ table.

bidirectional table (also accepted: crosstab or pivot table).

- (iv) The ratio of a data region's width to its height is known as the _____.
aspect ratio (width ÷ height). Decreasing it increases the apparent degree of change

Q3. Match the relationship in Column A with the most appropriate graphical solution in Column B. [2 Marks]

Column A — Relationship	Column B — Graphical Solution
(a) Correlation	(1) Box plot
(b) Part-to-whole	(2) Points only, with no connecting line
(c) Distribution across many groups	(3) Scatter plot with a trend line
(d) Time series with irregular intervals	(4) Unstacked bars, sorted by size, made to touch

Answer sequence: (a)–3, (b)–4, (c)–1, (d)–2.

Q4. State the conditions under which a table should be preferred over a graph, and briefly justify why each condition favours a table. [2 Marks]

Core statement: Tables make it easy to look up individual values. Use a table when one or more of the following is true —

Condition	Why a Table Serves It Best
1. The display will be used to look up individual values.	Text is understood directly, without any visual translation.
2. It will be used to compare individual values, but not the entire series of values.	The eye can go straight to two numbers; whole-series patterns are not required.
3. Precise values are required.	A table can carry a value such as 27.387483, which no visual encoding can express.
4. The quantitative information involves more than one unit of measure.	A graph normally carries a single quantitative scale in a single unit; a table can place units, dollars and percentages side by side.
5. Both summary and detail values are included.	A table can hold monthly details and an annual total in one display.

- Note: when the message is only one or two numbers, neither a table nor a graph is right — a sentence or bullet does the job (“Q2 sales = \$1,485,393, exceeding forecast by 16%”). There is eloquence in simplicity.

Marking: 1½ marks for any 3 correct conditions, 1½ marks for the justifications. Full 3 marks if all five conditions are listed with reasons.

Q5. Stephen Few recommends abandoning the pie chart altogether. Justify this recommendation and name the display that should replace it. [2 Marks]

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Q6. Context-based question. [2 Marks]

A monthly sales report carries a line graph titled “Sales Are Skyrocketing!”. The underlying data show that sales moved from \$19,500,000 in July to \$19,560,000 in December — an increase of less than one-third of one percent. On the page, however, the line appears to climb steeply and run off the top of the graph, suggesting growth of more than 200%.

Identify the design devices that produced this distortion and state the correction for each.

Principle violated: Maintain visual correspondence to quantity. The goal of design for communication is always to promote an accurate understanding of the truth. Five devices produced the distortion —

Design Device Used	Correction
1. The Y-axis scale does not start at zero; it runs only from \$19,475,000 to \$19,560,000, so minor changes appear extreme.	Include zero in the quantitative scale. If a narrowed scale is genuinely needed to reveal small differences among large values, alert readers explicitly with a note.
2. The plot area is taller than it is wide, dramatically increasing the slope of the line.	Never manipulate the aspect ratio to exaggerate change. By convention, make time-series graphs wider than they are tall.
3. The line is bright green on black — the colour of growth, health and dollars in English-speaking cultures — so it pops with visual impact.	Use muted, neutral colours for data that should not be highlighted; reserve bright, fully saturated hues for genuine emphasis.
4. The highest value (\$19,560,001) is set as the top of the scale, so the line appears to run off the top of the graph.	Set the top of the scale above the highest value so the data sits within the region.
5. The boldfaced axis label “Millions” sits in the prominent upper-left position beside the title, suggesting increases of millions.	Place and weight labels so that prominence matches importance; do not let a supporting label imply a magnitude the data does not contain.

- An even worse version of the same graph removes all Y-axis values and labels only the final data point, leaving no way whatsoever to judge the extent of the increase. Rendering it in 3-D and manipulating the viewing angle would exaggerate it further.
- **Honest redesign:** a zero-based line graph with a conventional wide aspect ratio, muted line colour, and a note stating the actual change — a rise of less than one-third of one percent over six months.

Marking: ½ mark per device correctly identified (2½ marks), 1½ marks for the corrections. Deduct nothing for order.

Q7. The design of every graph is constrained by how the eye and brain actually work.
[2 + 2 = 4 Marks]

- (a) Distinguish between preattentive and attentive processing, with one example of each.
- (b) List the visual attributes available for encoding data, state which of them are perceived quantitatively, and give the design rule that follows from this.

Q7 (a). Preattentive versus attentive processing [2 Marks]

Preattentive Processing	Attentive Processing
Occurs in iconic memory (the visual sensory register), which holds each snapshot for less than a second.	Occurs in working memory, which is temporary and limited to three or four chunks at a time.
Automatic, unconscious, parallel and extremely fast — recognition and nothing more.	Conscious, sequential and slow, but capable of learning and understanding.
Example: print the 5s in black among light-grey digits and they appear instantly.	Example: counting the 5s in four rows of uniformly grey digits takes several seconds.

- Design consequence: if you want something to stand out, encode it with a preattentive attribute that contrasts with its surroundings. If you want objects to be seen as a group, give them the same preattentive attribute.
- Preattentive processing usually cannot handle more than one visual attribute at a time — picking out “all black circles” is easy, but picking out “dark grey squares” when both shape and intensity vary requires slow attentive processing.

Q7 (b). Which attributes encode quantity [2 Marks]

Attribute (grouped by category)	Quantitatively Perceived?
Form — Length	Yes
Form — Width	Yes, but limited
Form — Size	Yes, but limited
Form — Orientation, Shape, Enclosure	No
Colour — Hue	No
Colour — Intensity (saturation and lightness)	Yes, but limited
Spatial position — 2-D position	Yes

- **The design rule:** whenever you can use length or 2-D position to encode quantitative values, do so. All of the attributes above can encode categorical differences, but only length and 2-D position encode quantity accurately.
- Why not area — given a small circle worth 1, answers for the large circle (worth 16) range from 5 to 50. We can tell one area is bigger, not by how much.
- Why not colour intensity — we can tell one object is darker provided the difference is large enough, but we cannot assign a value to the difference.
- Why not hue — different hues are perceived only as categorically different, never as more or less.
- Practical limits: no more than about eight distinguishable hues, about four orientations and about four sizes. Beyond that, readers fall back on slow attentive processing and then hit the three-to-four-chunk limit of working memory anyway.
- **Resulting encoding hierarchy:** (1) 2-D position — best for both quantities and categories; (2) length with a common baseline, requiring a zero base; (3) hue for categorical items only; (4) colour intensity, size and width for quantities only as a last resort, e.g. on maps; (5) shape, orientation, enclosure, fill pattern and line style for categories where colour is unavailable.

Marking: (a) 1 mark for the distinction, 1 mark for examples or design consequence. (b) 2 marks for the attribute table, 1 mark for the rule and its justification.